



North South University

Department of Electrical & Computer Engineering

Project Report

Project Title: Cloud-based Self-service Ordering System Using Arduino (Kiosk)

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Cloud-based Self-service Ordering Kiosk using Arduino

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Abstract—Technology gap exists in underdeveloped countries which can be compensated using on-budget projects. This paper aims to develop a product which will fill this gap in kiosk market at a minimum cost of price. A study on the latest kiosk technology is done, and the possible features which can be included on budget are identified and implemented on this project. Finally, the paper shows the use of Arduino, NodeMCU, Thin-Film Transistor(TFT) display, and Digital Logic Design (DLD) components.

Index Terms—Kiosk, Cloud, Arduino Mega, Firebase, TFT display, Digital logic design, TFT Algorithm, Cloud Algorithm.

I. INTRODUCTION

A self-service ordering kiosk allows a customer to interact with a company without any employer being involved. A kiosk's features can determine its different uses: ordinary and extraordinary. Some of the ordinary uses include filling out job applications, requesting appointments, submitting feedback, paying a bill, ticket printing, etc. [7] The latest uses may include accessing the internet, storing information on cloud, etc. [8] Various applications of a self service kiosk in shown in Fig 1.

Kiosks can be of two types: Interactive and Non-Interactive. Various shapes and sizes are available. A few examples are shown in Fig 2. Studies have shown that customers spend 30 percent more on self-service kiosks.[9] A proper self-service kiosk displays tempting images, compelling product descriptions, and easy-to-add modifiers so that it is much easier for customers to order what they desire.

Significance of this machine in the modern world, In the recent self-service economy where advancement in technology and automation is increasing and will continue to increase, Kiosks will play a part in it. As it is a labor and time saving tech and the free resources can be used elsewhere.

Alternatives for kiosk, the alternatives for kiosk in general sense are conventional store-system and online delivery system. In some sense kiosk is a better alternative and, in some sense, not so. In the growing automation dependent world kiosk is the future. It saves resources and manpower which now can be used effectively elsewhere. And to talk about the cons of kiosk it would be that a number of people will lose their jobs (mostly unskilled labor) and it maybe the end of many family-run businesses. Current Market Status, the global kiosk market

size is growing. The market is projected to reach USD 26940 million by 2026. In 2020 it was 18570 million that an estimate of 6.4 percentage CAGR(compound annual growth rate) during the period of 2021-2026.[1] The typical price range of self-service kiosk ranges from 1,500to5,000. Although this varies depending on various factors like Device Hardware, kiosk Hardware, Application, Kiosk system software etc. The more accessories or features is added the price range rises. [2] In the context of Bangladesh this price range is huge and taking into account that we don't manufacture kiosks ourselves there is an added import charge. So, a cheap solution is badly needed to keep up with today's automation heavy market. Although the installation of kiosks would mean that a number of people will lose their jobs but they are mostly cheap expandable labor. Also, the free resources now can be used elsewhere. Also, in the current pandemic situation where minimal human interaction is necessary, kiosks will play a vital role achieving that and more. That is why we decided on this project. Our purpose is to make prototype of a low-cost kiosk with cheap easily accessible maintenance and we used basic microcontroller, microprocessor and logic circuitry. Our project will also have a cloud data storage part which keep an account of all order data for later analysis of which items are more popular compared to other and many more.

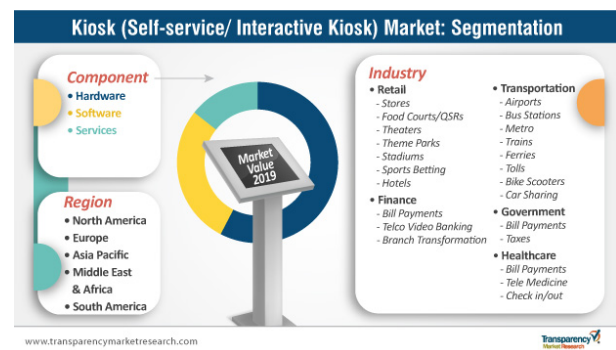


Fig. 1. Source: <https://transparencymarketresearch.com>

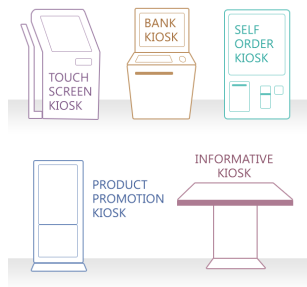


Fig. 2. Source: <https://pinterest.com>

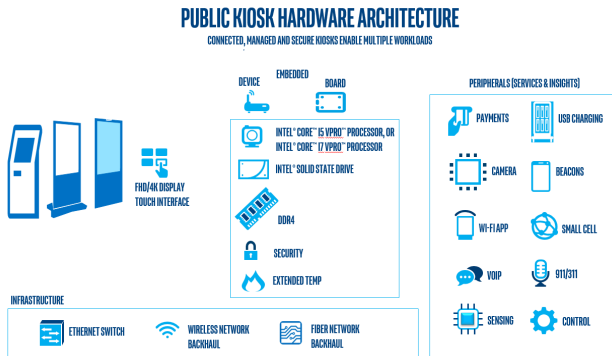


Fig. 3. Source: <https://software.intel.com>

II. LITERATURE REVIEW

The self-service technology began with the vending machines dating back to the 1800s. Those were small stands that offered cheap goods but it was the first time the customer was able to buy products without the help of a vender. The first self-service interactive kiosk was installed in the 1980s, the “Plato Hotline”, which was a university campus information-based system. [4] [5] In today’s world kiosks are almost used in every aspect from food-service, ATM, airline ticket vending kiosks and so on. And it’s not slowing down, in today’s self-service economy as automation increases kiosks will help save labor and resources. But the main focus is now to make kiosks cheaper as generally they tend to be very expensive with high maintenance. A lot of research is being done keeping this in mind. Like the Dickinson College Library with their library activity highlight kiosk developed by User Services division. They managed to get this done relatively cheap. For displayed information they used InfoPanelController.exe to control flow of page Interop.SHDocVw.dll, panelid.txt for other necessary features. And for data storage they used panelupdater.cmf for ID verification, getinfopaneldata.cmf for storing. [3] Another section of self-service technology is the analytical utilization of user data to give the customers the best experience. In a paper from KioskMasterplace.com there is the mention of standard way of collecting data by following three steps. The collection step includes using the right software and data storage for data storage and the data includes touches, length of transaction, usage of specific components etc. The reporting step includes

compiling the gathered data and make summary of it. And this has to be general, not individual. The best solution is a good cloud-based remote monitoring system with customizing reporting tools. The analytics step is to analyze and look for pattern in the data and gather useful insights. [4]

This project would be easier to make using a Raspberry Pi, since conventional operating systems (such as Linux) can be on it. And therefore, slideshow applications can be used for displaying the kiosk items that can easily be installed on the operating system (such as LibreOffice for Linux), instead of making a complete UI from scratch which had to be done in the Arduino. Raspberry Pi is also much more flexible for accessing files in the memory card, compared to the Arduino where we had to write multiple lines of codes for fetching the pictures of the items. There is also another advantage, as there is more firebase support for Raspberry Pi than for Arduino.[5]

Raspberry Pi can also run browsers unlike Arduino, so the whole kiosk model can be replaced with a web application and run it on the Raspberry Pi which will make the whole project much easier, compact and decrease the difficulty of maintenance. It also has a built-in kiosk-mode which can remove the cursor, and restrict the user from minimizing the application giving the whole system a genuine kiosk environment.

III. UNIQUE CONTRIBUTION

The main purpose of our project was to build a working prototype of a cheap self-service kiosk with easily available components. The cost factor was our main focus point while providing the necessary and latest features. In the global market, a latest Kiosk’s price ranges between 120k to 400k Taka. If we integrate our project to an actual device, the estimated total cost would still be within 40,000 Taka even with a robust body frame. Raspberry pi would be used in our final product instead of Arduino, which opens up opportunities for new features to be added, as well as for using cloud stored information for further data analysis. This removes any limitations of Arduino, and gives our product the potential to work as efficiently as the latest kiosks. Moreover, the components used to make out device is easily available and inexpensive so there is an additional advantage of local maintenance.

IV. METHODS

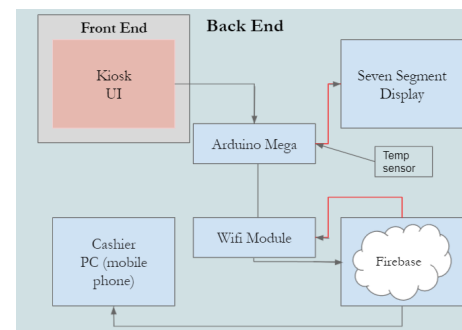


Fig. 4. Full block diagram of Prototype

The main concept of this project is to collect data of order from the kiosk, display the token number on the seven segment display, store the data to firebase and finally display the data to the cashier. Fig. 4 shows the overall system starting from Kiosk UI and ending in Cashier PC and Seven Segment Display.

A. Hardware Implementation

List of Components used: IL9486 TFT display, Arduino Mega, Arduino Uno, NodeMCU, LM35 Temperature sensor, voltage regulator, 7474 IC as flipflop, 74151 IC as mux, 7 segment display (common anode), Led light, jumper wire, resistor (440ohm) and breadboard.

Here, a circuit is built that will take 3-bit input from the Arduino and display the output in a 7-segment display. We have used 7 mux and the inputs from Arduino were used as a select bit for all the mux. Then, the input of the mux is manually given using VCC as 1 and Ground as 0. We set all the mux in Ascending order from A to G. Then, we used the outputs of the muxes as inputs of the 7-segment display.

A	B	C	A [*]	B [*]	C [*]	D _A	D _B	D _C
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0

Fig. 5. Truth Table for Sequential Circuit

We are using asynchronous model to build a 3 bit up counter using d-flip flops.here is clock diagrams.

The name Data Latch refers to a D Type flip-flop that is level triggered, as the data (1 or 0) appearing at D can be held or 'latched' at any time whilst the CLK input is at a high level if the data at D changes during this time, the Q output assumes the same logic level as the D.

if there are changes in the data during period when the clock pulse is at its high level, the logic state at Q changes in sympathy with D, and only 'remembers' the last input state that occurred during the clock pulse This effect is called 'Ripple Through', and although this allows the level triggered D Type flip-flop to be used as a data switch, only allowing data through from D to Q as long as CLK is held at logic 1, this may not be a desirable property in many types of circuit.

As we can see, the combinational circuits we have made shows number(1-8) in the 7-segment display. Here we use 8:1

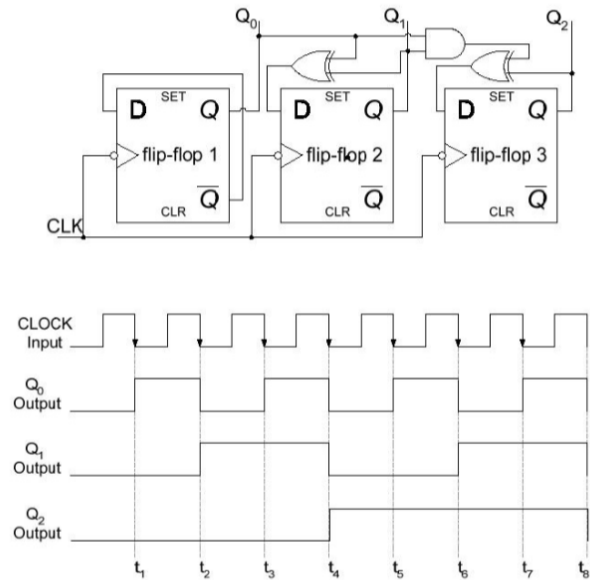


Fig. 6. Sequential Circuit Diagram

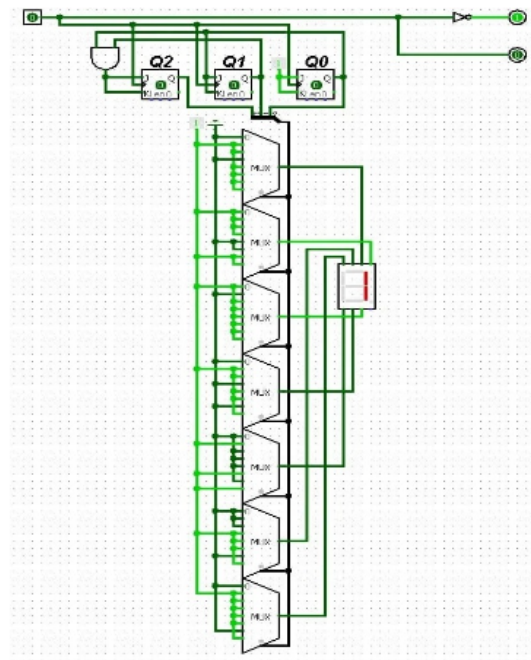


Fig. 7. Full Sequential Circuit Diagram

mux, and use D flipflop. Next we have to connect circuit with the Arduino ,input pin with blunk and connect vcc, and ground pin . Then we get output sequentially.

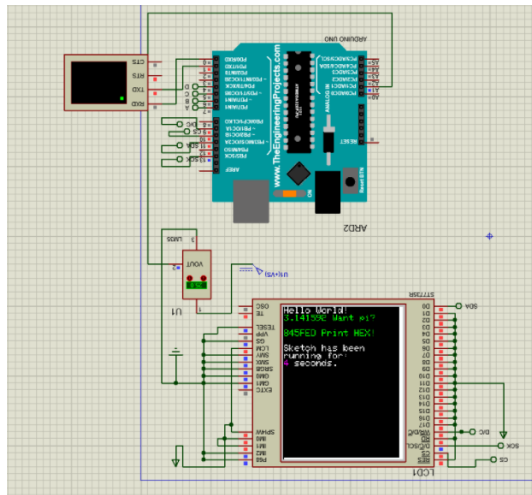


Fig. 8. Connection of Temperature Sensor and a TFT display(ST7735) with Arduino UNO in Proteus.

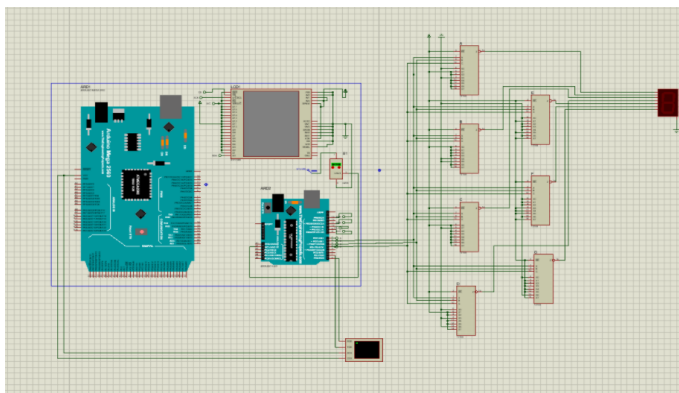


Fig. 9. Full Circuit in Proteus.

Steps for menu creation:

- Necessary libraries has been included.
- In section 1, string function was created using library-made functions. This new function contains string, x and y coordinates, front color, black color, and mode of text.
- In section 4, the five menu category functions were created for Main Dish, Appetizer, Kids menu, Dessert and Drinks, in which the string function created in section 1 was used.
- In section 5, the initial look of Kiosk interface was created which includes the "Place order", "Cart", "Add items" and "remove items" buttons, and the order status. It also includes the name of the menu categories. This section is done under setup which runs only once.
- In section 6a, the code decides whether Main Dish menu is selected or unselected when the xy plane for Main Dish menu is pressed. The code differentiates between select and unselect using the "switch flag 6" boolean, which is false for selected and true for latter.

When selected: The color of the Main Dish category becomes white, while all other categories remain the same color. Area between $y=147$ and $y=250$ becomes white and the Main dish menu appears there as the Main dish function created in section 4 is called.

- In section 6b: When the switch flag is false i.e., the main dish is selected, each food item from the main dish menu can be selected by pressing the coordinate range allocated for them. One food item can be selected at a time and the allocated area for that appears blue. Only the picture of the food item selected is shown in the image area (between $x=180$ and $x=320$, and between $y=38$ and $y=138$). Here, ssn (where n is 1,2,3,4,5) boolean is used to differentiate between the selected item and unselected item.

- In section 6c, the ingredients are defined for the selected food time.

- In section 6d, the number of items in Main Dish is counted for displaying in the cart. sn (where $n=1,2,3,4,5$) variables were defined as 0 at the beginning of the code. If else loop was used here. If add items button created in section 5 is pressed, item is added so sn gets added by 1. If remove items button from section 5 is pressed, 1 item is removed so sn gets subtracted by 1.

- Section 7, 8,9 and 10 were carried out the same way as section 6 but for different menu categories. 7 for Appetizer menu; 8 for Kids menu; 9 for Dessert menu; 10 for Drinks menu.

- In section 11, If else loop is used to recommend item using temperature sensor data. If the temperature went above 30 it displays advertisement for Apple Juice, otherwise it advertises Hot coffee.

- In section 12: The cart is designed using different functions from library and section 1. If-else and while loops are used to align the items added to the cart. m variable is used for horizontal shifting of the food items in the cart.

- In section 13: This section deals with placing order. Functions like string function and draw color function is used to change the status of order. The token number is added by 1 every time an order is placed and all the items in the cart gets removed i.e., for example, for main dish items all sn variables become zero again.

- In section 14: This section deals with printing information. `printer.print` inbuilt function is used to print the order list details. `printer.setSize` is used for resizing the fonts. if loop is used here to ensure that only the selected food items are printed. The printer is put to sleep for three seconds and woken up before printing again.

- In section 15: Input is sent to the sequential circuit using pin 31 which acts as a clock.

- In section 16: Here, the order information is sent to the NodeMCU as one string. For this, all the variables were converted to strings.

Steps for display Picture:

Image converter from UTFT library is used. Here we used flash memory of our display.

- The image is resized to the dimension of the allocated area used in the display. The picture is converted to R5G6B5 color as suited for our IL9486 display.
- The image is saved as 16 bit color using paint and converted to Bitmap which converts picture to ASCII characters using UTFT library image converter. The bitmap is then saved as a c file. The c file is saved in the same folder of Arduino files as our kiosk file (shown in Fig. 16 in Appendix).
- The images are included in the beginning of the code as “imagenam.c”. One .c file is included for each food item.
- In section 2, functions are created for the images which is later used for image showing.

B. Data Transfer

To bring into account the cloud storage part of our project, the extra components that we needed were the NodeMCU and the Android studio which is an integrated development environment for android OS. To briefly explain step by step the whole process with the help of block diagram in figure 10,

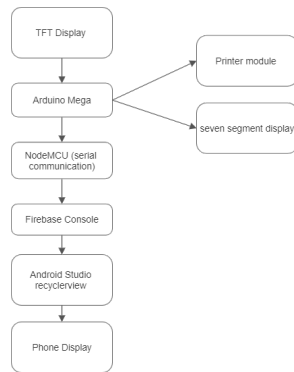


Fig. 10.

Sending of user data to firebase: To connect NodeMCU with Arduino Mega, Arduino's pin 18 and 19(Analog communication pin) were connected to NodeMCU's Fx (GPIO03) and Tx (GPIO01) pin, now these two components can be regarded as one. For coding, necessary header files were added, firebase host was defined, and authenticator was connected with the designated database and Wi-Fi SSID was defined and powered to connect to the local Wi-Fi. Then in main body code, connection Firebase.Bagin(Host,Authenticator) was initialized. Also, in this part connection to local Wi-Fi was configured and the status displayed as 'connected'. Then in the loop function, initial value, handle error, update value were set, and then finally string was set to send to the firebase database.

C. Data Receive

For the data receive part, Android Studio was used. Basically, a small app was made which gets the data from Firebase and displays it on cell phone as the built-in emulator was not used. The cell phone here, in the context of our project, is for the cashier end. Then the Firebase website's Instructions of data receive from firebase was followed. To briefly specify the whole process, java was used and four files were created.

item.xml and activityuserlist.xml for the layout. Here, xml was used to create a linear layout of the data in the receive monitor section. Then two java class to specify which data are to be fetched. And lastly MyAdapter.java for connecting Android Studio to the Firebase data-base. At the very first, firebase was connected to android-studio and dependences were added. Then in UselistActivity, layout resource file named item, a layout was created for the recycler view. Then in the uselistactivity a class for the user data orders and token was created. Then, an adapter class and extend recycler view in it were added. Then in the userlist class, a reference variable was created. The main advantage of this data storage part is that now from our knowledge from literature review by following some well-established steps we can analyze these data and give customers the best experience.

V. RESULTS

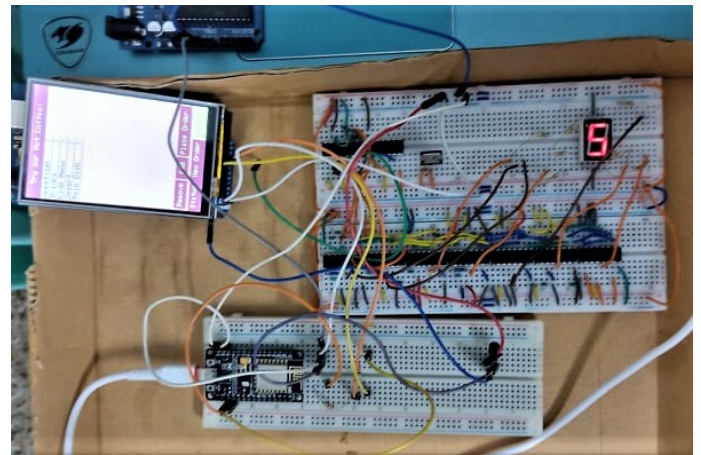


Fig. 11. Full Project Circuit



Fig. 12. Kiosk UI



Fig. 13. Kiosk Cart



Fig. 14. Firebase database

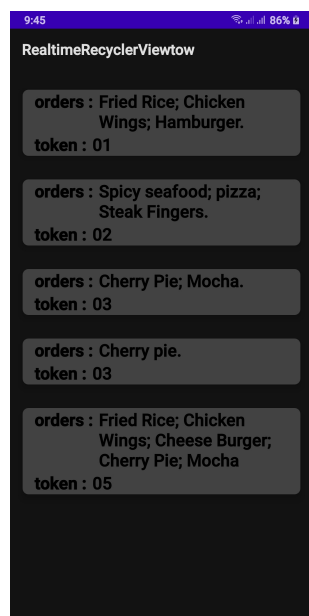


Fig. 15. Data receive monitor

VI. DISCUSSION

Comparatively self-service tech especially kiosks in food service system in our country is not very much used. Kiosks of the lower price end like Information Kiosk TSK-1000 goes for around 56,000tk and high-end kiosks like 43inch Touch Monitor Kiosk FT43TMKCAPOB goes for 1.69Lakh. Our basic prototype cost us in total 8,000tk and even if we integrate this into an actual device using Raspberry pi, Raspberry pi 7" touch display, our own Cloud storage and more, it will cost us no more than 40,000 tk. Also, the maintenance cost should be relatively cheap as all the components are easily available.

In retrieving data from Firebase data base, our original plan was to make a user interface for the receive part but because of shortage of time we were just able to retrieve the data from data base and display the data in our phone console. Moreover, we were unable to show proper alignment of food items in the cart, as well as to show all the images of the food items due to limitation of flash memory of Arduino Mega. So, for future improvement these can be the main goals: to make a more working user interactive UI in data display part and analyze data for research; to make a better organized Kiosk UI. Lastly, our kiosk UI code could be made dry(by building new functions) to avoid redundancy.

VII. CONCLUSION

As we have seen, we were able to build an economical Kiosk prototype with easily available components. The future improvements of our project will mainly focus and work with the user data that we will be able to collect. Using data analysis, we can make it beneficial for restaurant market, and with the help of additional features, we can make our kiosk more user-friendly and serviceable to the customers.

VIII. REFERENCES

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- [8] "RSI Geeks," 18 November 2020. [Online]. Available: <https://rsigeeks.com/blog/7-features-of-self-service-kiosks/>. [Accessed 19 September 2021].
- [9] "Self-Service Kiosks Boost Consumer Spending By 30 Percent," PYMNTS, 26 June 2018.

IX. APPENDICES

A. Android Studio Code

```
//Activity_main.xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity 01"
        android:layout_centerVertical="true"
        android:layout_centerHorizontal="true"
        android:id="@+id/textView"/>

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="open activity 02"
        android:layout_below="@+id/textView"
        android:layout_centerHorizontal="true"/>

</RelativeLayout>

//MainActivity.java
package com.example.realtimerecyclerviewtow;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
    private Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        button = (Button) findViewById(R.id.button);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                openuserlist();
            }
        });
    }

    public void openuserlist(){
        Intent intent= new Intent(this, userlist.class);
        startActivity(intent);
    }
}

//activity_userlist.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".userlist">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/userList"
        android:layout_width="409dp"
        android:layout_height="729dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

Item.xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.cardview.widget.CardView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    app:cardElevation="8dp"
    app:cardCornerRadius="8dp"
    android:layout_margin="16dp">

    <LinearLayout
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginLeft="16dp"
        android:orientation="vertical">

        <LinearLayout
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_gravity="center_vertical"
            android:orientation="horizontal">

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
```



```

        android:text="orders : "
        android:textColor="@color/black"
        android:textSize="26sp"
        android:textStyle="bold"/>
<TextView
    android:id="@+id/tvorders"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginLeft="8dp"
    android:text="chicken burger"
    android:textColor="@color/black"
    android:textSize="26sp" />

</LinearLayout>
<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_vertical"
    android:orientation="horizontal">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="token : "
        android:textColor="@color/black"
        android:textSize="26sp"
        android:textStyle="bold"/>

    <TextView
        android:id="@+id/tvtoken"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginLeft="8dp"
        android:text="01"
        android:textColor="@color/black"
        android:textSize="26sp" />

</LinearLayout>

</LinearLayout>

</androidx.cardview.widget.CardView>

```

```

Userlist.java
package com.example.realtimerecyclerviewtow;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import android.os.Bundle;

import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import java.util.ArrayList;

public class userlist extends AppCompatActivity {

    RecyclerView recycler View;
    DatabaseReference database;
    MyAdapter my Adapter;
    ArrayList<User> list;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_userlist);

        recyclerView = findViewById(R.id.userList);
        database = FirebaseDatabase.getInstance().getReference("Users");
        recyclerView.setHasFixedSize(true);
        recyclerView.setLayoutManager(new LinearLayoutManager(this));

        list = new ArrayList<>();
        myAdapter= new MyAdapter(this,list);
        recyclerView.setAdapter(myAdapter);

        database.addValueEventListener(new ValueEventListener() {
            @Override
            public void onDataChange(@NonNull DataSnapshot snapshot) {
                for (DataSnapshot dataSnapshot : snapshot.getChildren()){
                    User user = dataSnapshot.getValue(User.class);
                    list.add(user);
                }

                myAdapter.notifyDataSetChanged();
            }
        })

        @Override
        public void onCancelled(@NonNull DatabaseError error) {
        }
    }
}

```

```

    });

}

}

MyAdapter.java
package com.example.realtimerecyclerviewtow;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.ArrayList;

public class MyAdapter extends RecyclerView.Adapter<MyAdapter.MyViewHolder> {

    Context context;

    ArrayList<User> list;

    public MyAdapter(Context context, ArrayList<User> list) {
        this.context = context;
        this.list = list;
    }

    @NonNull
    @Override
    public MyViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        View v = LayoutInflater.from(context).inflate(R.layout.item,parent,false);
        return new MyViewHolder(v);
    }

    @Override
    public void onBindViewHolder(@NonNull MyViewHolder holder, int position) {

        User user = list.get(position);
        holder.orders.setText(user.getOrders());
        holder.token.setText(user.getToken());

    }

    @Override
    public int getItemCount() {
        return list.size();
    }

    public static class MyViewHolder extends RecyclerView.ViewHolder{

        TextView orders, token;

        public MyViewHolder(@NonNull View itemView) {
            super(itemView);

            orders = itemView.findViewById(R.id.tvorders);
            token = itemView.findViewById(R.id.tvtoken);

        }
    }
}

```

B. Arduino Code

```
//Temperature Sensor variable
int val;

//Libraries used
#include "Adafruit_Thermal.h"
#include "SoftwareSerial.h"
#include <TouchScreen.h> //touch library
#include <LCDWIKI_GUI.h> //Core graphics library
#include <LCDWIKI_KBV.h> //Hardware-specific library
#include <Adafruit_GFX.h> // Core graphics library
#include <Adafruit_TFTLCD.h> // Hardware-specific library
#include <MCUFRIEND_kbv.h>
MCUFRIEND_kbv tft;

//All images used using Flash memory
#include "jpg.c"
#include "button.c"
#include "images.c"
#include "font.h"
#include "16-bitFR.c"
#include "Spicey_seafood.c"
#include "spaghettitwo.c"
#include "diffbutton.c"
#include "buttonlast.c"

//For Serial Transmission from Arduino Mega to NodeMCU
#include <SoftwareSerial.h>
SoftwareSerial espSerial(39, 41);
String str;

LCDWIKI_KBV my_lcd (ILI9486,A3,A2,A1,A0,A4); //model,cs,cd,wr,rd,reset

//Defining Colors in hexadecimal
#define BLACK 0x0000
#define BLUE 0x001F
#define RED 0xF800
#define GREEN 0x07E0
#define CYAN 0x07FF
#define MAGENTA 0xF81F
#define YELLOW 0xFFE0
#define WHITE 0xFFFF
#define CREAM 0xFFDDDD
#define DARKRED 0x990000
#define LIGHTYELLOW 0xFFFFCC
#define GREY 0xA9A9A9

//printer
#define TX_PIN 2
#define RX_PIN 3

SoftwareSerial mySerial(RX_PIN, TX_PIN)
Adafruit_Thermal printer(&mySerial);

char *color_name[] = { "BLUE", "GREEN", "RED", "WHITE", "CYAN", "MAGENTA", "YELLOW" };
uint16_t color_mask[] = { 0x001F, 0x07E0, 0xF800, 0xFFFF, 0x07FF, 0xF81F, 0xFFE0 };

#define YP A3 // must be an analog pin, use "An" notation!
#define XM A2 // must be an analog pin, use "An" notation!

#define YM 9 // can be a digital pin
#define XP 8 // can be a digital pin

#define TS_MINX 906
#define TS_MAXX 116

#define TS_MINY 92
#define TS_MAXY 952

#define STATUS_X 10
#define STATUS_Y 65

boolean switch_flag_2 = true, switch_flag_3 = true, switch_flag_4 = true, switch_flag_5 = true, switch_flag_6 = true, switch_flag_cart = true;
int16_t menu_flag = 1;
boolean menu_flag1 = true, menu_flag2 = true, menu_flag3 = true, menu_flag4 = true, menu_flag5 = true;
boolean cart = false;
boolean ss1 = true, ss2 = true, ss3 = true, ss4 = true, ss5 = true;
boolean dd1, dd2, dd3, dd4, dd5 = true;
boolean ff1, ff2, ff3, ff4, ff5 = true;
boolean gg1, gg2, gg3, gg4, gg5 = true;
boolean hh1, hh2, hh3, hh4, hh5 = true;

int i=0, j=0, k=0;
int token=1;

//Menu Dish Item Numbers in Cart
int s1=0, s2=0, s3=0, s4=0, s5=0;
//Appetizer Item Numbers in Cart
int d1=0, d2=0, d3=0, d4=0, d5=0;
//Kids menu Item Numbers in Cart
int f1=0, f2=0, f3=0, f4=0, f5=0;
//Dessert Item Numbers in Cart
int g1=0, g2=0, g3=0, g4=0, g5=0;
//Drinks Item Numbers in Cart
int h1=0, h2=0, h3=0, h4=0, h5=0;

// Cart y axis shifting
int m=0;

TouchScreen ts = TouchScreen(XP, YP, XM, YM, 300);
```

```

//Section 1: Function for displaying string
void show_string(uint8_t *str,int16_t x,int16_t y,uint8_t csize,uint16_t fc, uint16_t bc,boolean mode)
{
    my_lcd.Print_String(str,x,y);
    my_lcd.Set_Text_Size(csize);
    my_lcd.Set_Text_colour(fc);
    my_lcd.Set_Text_Back_colour(bc);
    my_lcd.Set_Text_Mode(mode);
}

//Section 2: Function for displaying pictures of food items
void show_pic(int16_t x1,int16_t y1,int16_t x2,int16_t y2)
{
    my_lcd.Set_Addr_Window(x1, y1, x2, y2);
    my_lcd.Push_Any_Color(buttonlast, 0xE4, 1, 1);
}
void show_pic(void)
{
    my_lcd.Set_Addr_Window(my_lcd.Get_Display_Width()-140, 38, my_lcd.Get_Display_Width(), 138 );
    my_lcd.Push_Any_Color(fried_rice2, 14000, 1, 1);
}
void show_pic2(void)
{
    my_lcd.Set_Addr_Window(my_lcd.Get_Display_Width()-140, 38, my_lcd.Get_Display_Width(), 138 );
    my_lcd.Push_Any_Color(Jjampong, 14000, 1, 1);
}

void show_pic3(void)
{
    my_lcd.Set_Addr_Window(my_lcd.Get_Display_Width()-140, 38, my_lcd.Get_Display_Width(), 138 );
    my_lcd.Push_Any_Color(fried_rice2, 14000, 1, 1);
}

//for scrolling: Did not work
void windowScroll(int16_t x, int16_t y, int16_t wid, int16_t ht, int16_t dx, int16_t dy, uint16_t *buf)
{
    if (dx)
    {
        for (int16_t row = 0; row < ht; row++)
        {
            my_lcd.ReadGRAM(x, y + row, buf,wid, 1);
            my_lcd.Set_Addr_Window(x, y + row, x + wid - 1, y + row);
            my_lcd.Push_Any_Color(buf + dx, wid - dx, 1,0);
            my_lcd.Push_Any_Color(buf + 0, dx, 0,0);
        }
    }
    if (dy)
    {
        for (int16_t col = 0; col < wid; col++)
        {
            my_lcd.ReadGRAM(x + col, y, buf,1, ht);
            my_lcd.Set_Addr_Window(x + col, y, x + col, y + ht - 1);
            my_lcd.Push_Any_Color(buf + dy, ht - dy, 1,0);
            my_lcd.Push_Any_Color(buf + 0, dy, 0,0);
        }
    }
}

//Section 3: Check whether pressed or not
boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
{
    if((px > x1 && px < x2) && (py > y1 && py < y2))
    {
        return true;
    }
    else
    {
        return false;
    }
}

//Section 4: Catagorizing each menu
void MainDish_menu(void)
{
    show_string("Fried Rice      120TK",13,150,2,BLACK, BLACK,1);
    show_string("Spicey seafood  200TK",13,170,2,BLACK, BLACK,1);
    show_string("Pizza          700TK",13,190,2,BLACK, BLACK,1);
    show_string("Spaghetti      180TK",13,210,2,BLACK, BLACK,1);
    show_string("Pasta          170TK",13,230,2,BLACK, BLACK,1);
}

void Appetizer_menu(void)
{
    show_string("Chicken Wings  150TK",10,150,2,BLACK, BLACK,1);
    show_string("Pizza Fries   150TK",10,170,2,BLACK, BLACK,1);
    show_string("Rice Ball     120TK",10,190,2,BLACK, BLACK,1);
    show_string("Beef Patty    100TK",10,210,2,BLACK, BLACK,1);
    show_string("Broccoli Rabe  70TK",10,230,2,BLACK, BLACK,1);
}

void Kids_menu(void)
{
    show_string("Cheese Sandwich 100TK",10,150,2,BLACK, BLACK,1);
    show_string("Hamburger      140TK",10,170,2,BLACK, BLACK,1);
    show_string("Steak Fingers  120TK",10,190,2,BLACK, BLACK,1);
    show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);
    show_string("Beef Steak     160TK",10,230,2,BLACK, BLACK,1);
}

void Dessert_menu(void)
{

```

```

        show_string("Lava Cake          120TK",10,150,2,BLACK, BLACK,1);
        show_string("Cheesecake        130TK",10,170,2,BLACK, BLACK,1);
        show_string("Maple Apple Pie   130TK",10,190,2,BLACK, BLACK,1);
        show_string("Brownie Pops      90TK",10,210,2,BLACK, BLACK,1);
        show_string("Cherry pie        120TK",10,230,2,BLACK, BLACK,1);

    }

    void Drinks_menu(void)
    {
        show_string("Cappuccino        100TK",10,150,2,BLACK, BLACK,1);
        show_string("Glace              80TK",10,170,2,BLACK, BLACK,1);
        show_string("Latte              100TK",10,190,2,BLACK, BLACK,1);
        show_string("Cocoa              120TK",10,210,2,BLACK, BLACK,1);
        show_string("Mocha              120TK",10,230,2,BLACK, BLACK,1);

    }

    void setup(void)
    {
        Serial.begin(9600);
        my_lcd.Init_LCD();
        printer.begin();
        printer_orders();

        my_lcd.Fill_Screen(CREAM);
        my_lcd.Fill_Rectangle(0, 0, my_lcd.Get_Display_Width()-1, 480);
        show_string(" ",0.5*my_lcd.Get_Display_Width(),240,2,BLACK, BLACK,1);
        show_string("KIOSK",0.5*my_lcd.Get_Display_Width()-40,220,2,GREY, GREY,1);
        delay(500);

        my_lcd.Init_LCD();
        my_lcd.Fill_Screen(CREAM);

//Section 5: Interface Initial Look
//ORDER Status
        my_lcd.Set_Draw_color(GREY); //Lower part color
        my_lcd.Fill_Rectangle(0, 443, my_lcd.Get_Display_Width()-1, 480);
        show_string(" ",11,410,2,LIGHTYELLOW, LIGHTYELLOW,1);
        show_string("Status: New Order",10,450,2,LIGHTYELLOW, LIGHTYELLOW,1);

//CART
        my_lcd.Set_Draw_color(LIGHTYELLOW); //Lower part color
        my_lcd.Fill_Rectangle(220, 443, my_lcd.Get_Display_Width()-1, 480);
        show_string(" ",230,450,2,GREY, GREY,1);
        show_string("Cart ",230,450,2,GREY, GREY,1);
        show_string(" ",304,450,2,LIGHTYELLOW, LIGHTYELLOW,1);

//Remove item
        my_lcd.Set_Draw_color(GREY);
        my_lcd.Fill_Rectangle(5, 400, 90, 435);
        show_string(" ",11,410,2,LIGHTYELLOW, LIGHTYELLOW,1);
        show_string("Remove",11,410,2,LIGHTYELLOW, LIGHTYELLOW,1);

//Add item
        my_lcd.Set_Draw_color(GREY);
        my_lcd.Fill_Rectangle(95, 400, 160, 435);
        show_string("Add", 111,410,2,LIGHTYELLOW, LIGHTYELLOW,1);

//PLACE ORDER
        my_lcd.Set_Draw_color(GREY);
        my_lcd.Fill_Rectangle(my_lcd.Get_Display_Width()-155, 400,my_lcd.Get_Display_Width()-5, 435);
        show_string("Place Order",my_lcd.Get_Display_Width()-150,410,2,LIGHTYELLOW, LIGHTYELLOW,1);

//UI color: upper part
        my_lcd.Set_Draw_color(GREY); //Upper part color
        my_lcd.Fill_Rectangle(0, 0, my_lcd.Get_Display_Width()-1, 37);

//Setting Horizontal line color
        my_lcd.Set_Draw_color(1, 1, 1); // grey lines

//Naming menu catagories
        my_lcd.Draw_Fast_HLine(0, 37, my_lcd.Get_Display_Width()); // (0,37) is the coordinates.
        show_string(" ",5,40,2,BLUE, BLACK,1);
        show_string("Appetizer >",5,40,2,BLUE, BLACK,1); // y=40

        my_lcd.Draw_Fast_HLine(0, 57, my_lcd.Get_Display_Width()-140);
        show_string("Drinks >",5,60,2,BLUE, BLACK,1);

        my_lcd.Draw_Fast_HLine(0, 77, my_lcd.Get_Display_Width()-140);
        show_string("Kids Menu >",5,80,2,BLUE, BLACK,1);

        my_lcd.Draw_Fast_HLine(0, 97, my_lcd.Get_Display_Width()-140);
        show_string("Desert >",5,100,2,BLUE, BLACK,1);

        my_lcd.Draw_Fast_HLine(0, 116, my_lcd.Get_Display_Width()-140);
        show_string("Main Dish >",5,119,2,BLUE, BLACK,1);

        my_lcd.Draw_Fast_HLine(0, 137, my_lcd.Get_Display_Width()-140);

//Serial Transmission
        espSerial.begin(115200);

        pinMode(31,OUTPUT);

    }

//Touch pressure defined
#define MINPRESSURE 10
#define MAXPRESSURE 1000

void loop(void)

```



```

{

    TSPoint p = ts.getPoint();

    pinMode(XM, OUTPUT);
    pinMode(YF, OUTPUT);
    if (p.z > MINPRESSURE && p.z < MAXPRESSURE)
    {
        p.x = map(p.x, TS_MINX, TS_MAXX, my_lcd.Get_Display_Width(),0);
        p.y = map(p.y, TS_MINY, TS_MAXY, my_lcd.Get_Display_Height(),0);

//Section 6a: Selecting and unselecting Main Dish by pressing
        if(is_pressed(5,119,my_lcd.Get_Display_Width()-1,137,p.x,p.y)) //boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
        {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);

            switch_flag_2=true;
            switch_flag_3=true;
            switch_flag_4=true;
            switch_flag_5=true;

            my_lcd.Set_Draw_color(CREAM);
            my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 137);
            show_string(" ",5,119,2,BLUE, BLACK,1);
            show_string("Appetizer >",5,40,2,BLUE, BLACK,1);
            show_string("Drinks >",5,60,2,BLUE, BLACK,1);
            show_string("Kids menu >",5,80,2,BLUE, BLACK,1);
            show_string("Desert >",5,100,2,BLUE, BLACK,1);
            show_string("Main Dish >",5,119,2,BLUE, BLACK,1);

            if(switch_flag_6)
            {
                my_lcd.Set_Draw_color(WHITE);
                my_lcd.Fill_Rectangle(0, 117, my_lcd.Get_Display_Width()-140, 137);
                show_string(" ",5,119,2,BLUE, BLACK,1);
                show_string("Main Dish <",5,119,2,BLUE, BLACK,1);

                MainDish_menu();
                switch_flag_6 = false; //Selected
            }
            else
            {
                my_lcd.Set_Draw_color(WHITE);
                my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);
                switch_flag_6 = true; //Not selected
            }
        }

//Section 6b: If Main Dish menu is selected:

        if(switch_flag_6==false)
        {

//Select Fried Rice and Unselect all
            if(is_pressed(0,147,my_lcd.Get_Display_Width()-1,166,p.x,p.y)){

                if(ss1)
                {
                    my_lcd.Set_Draw_color(BLUE);
                    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
                    show_string("Fried Rice 120TK",13,150,2,BLACK, BLACK,1);
                    ss1=false;
                    show_pic();
                    my_lcd.Set_Draw_color(LIGHTYELLOW);
                    my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

                    show_string(" ",10,270,1,BLACK, BLACK,1);
                    show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
                    show_string("Rice, egg, carrots, salt, pepper, soya sauce, olive oil,",10,290,1,BLACK, BLACK,1);
                    show_string("chicken",10,310,1,BLACK, BLACK,1);

                    my_lcd.Set_Draw_color(WHITE);
                    my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
                    show_string(" ",13,170,2,BLACK, BLACK,1);
                    show_string("Spicy seafood 200TK",13,170,2,BLACK, BLACK,1);
                    ss2 = true;

                    my_lcd.Set_Draw_color(WHITE);
                    my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
                    show_string("Pizza 700TK",13,190,2,BLACK, BLACK,1);
                    ss3 = true;

                    my_lcd.Set_Draw_color(WHITE);
                    my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
                    show_string("Spaghetti 180TK",13,210,2,BLACK, BLACK,1);
                    ss4 = true;

                    my_lcd.Set_Draw_color(WHITE);
                    my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
                    show_string("Pasta 170TK",13,230,2,BLACK, BLACK,1);
                    ss5 = true;

                }
                else {
                    my_lcd.Set_Draw_color(WHITE);
                    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
                    show_string("Fried Rice",13,150,2,BLACK, BLACK,1);
                    ss1 = true;
                }
            }

//Select Spicy food and Unselect all
            else if(is_pressed(0,167,my_lcd.Get_Display_Width()-1,186,p.x,p.y)){

```

```

if(ss2)
{
    my_lcd.Set_Draw_color(BLUE);
    my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
    show_string("Spicey seafood    200TK",10,170,2,BLACK, BLACK,1);
    ss2 = false;

    show_pic2();

    my_lcd.Set_Draw_color(LIGHTYELLOW);
    my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

    show_string(" ",10,270,1,BLACK, BLACK,1);
    show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
    show_string("Olive oil, Blub funnel, garlic cloves, salt, white wine",10,290,1,BLACK, BLACK,1);
    show_string("tomatoes, chicken, saffron, white fish, shrimp, orange zest",10,310,1,BLACK, BLACK,1);

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
    show_string(" ",10,150,2,BLACK, BLACK,1);
    show_string("Fried Rice        120TK",10,150,2,BLACK, BLACK,1);
    ss1 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
    show_string("Pizza            700TK",10,190,2,BLACK, BLACK,1);
    ss3 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
    show_string("Spaghetti        180TK",10,210,2,BLACK, BLACK,1);
    ss4 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
    show_string("Pasta            170TK",10,230,2,BLACK, BLACK,1);
    ss5 = true;
}
else {
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
    show_string("Spicey seafood    200TK",10,170,2,BLACK, BLACK,1);
    ss2=true;
}
}

//Select Pizza and Unselect all
else if (is_pressed(0,187,my_lcd.Get_Display_Width()-1,206,p.x,p.y)){

if(ss3)
{
    my_lcd.Set_Draw_color(BLUE);
    my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
    show_string("Pizza            700TK",10,190,2,BLACK, BLACK,1);

    ss3=false;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
    show_string("Fried Rice        120TK",10,150,2,BLACK, BLACK,1);
    ss1 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
    show_string("Spicey seafood    200TK",10,170,2,BLACK, BLACK,1);
    ss2 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
    show_string("Spaghetti        180TK",10,210,2,BLACK, BLACK,1);
    ss4 = true;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
    show_string("Pasta            170TK",10,230,2,BLACK, BLACK,1);
    ss5 = true;
}
else {
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
    show_string("Pizza            700TK",10,190,2,BLACK, BLACK,1);

    ss3=true;
}
}

//Select Spaghetti and Unselect all
else if (is_pressed(0,207,my_lcd.Get_Display_Width()-1,226,p.x,p.y)){

if(ss4)
{
    my_lcd.Set_Draw_color(BLUE);
    my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
    show_string("Spaghetti        180TK",10,210,2,BLACK, BLACK,1);
    show_pic3();
    ss4=false;

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
    show_string("Fried Rice        120TK",10,150,2,BLACK, BLACK,1);
    ss1 = true;
}
}

```

```

        my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Spicey seafood  200TK",10,170,2,BLACK, BLACK,1);
ss2 = true;

```

```

        my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Pizza          700TK",10,190,2,BLACK, BLACK,1);
ss3 = true;

```

```

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Pasta          170TK",10,230,2,BLACK, BLACK,1);
ss5 = true;
    }
    else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Spaghetti     180TK",10,210,2,BLACK, BLACK,1);

ss4=true;
    }
}

```

//Select Pasta and Unselect all

```

else if(is_pressed(0,227,my_lcd.Get_Display_Width()-1,246,p.x,p.y))
{if(ss5)
{
    my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Pasta          170TK",10,230,2,BLACK, BLACK,1);

    ss5=false;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Fried Rice     120TK",10,150,2,BLACK, BLACK,1);
    ss1 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Spicey seafood  200TK",10,170,2,BLACK, BLACK,1);
    ss2 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Pizza          700TK",10,190,2,BLACK, BLACK,1);
    ss3 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Spaghetti     180TK",10,210,2,BLACK, BLACK,1);
    ss4 = true;

    }
    else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Pasta          170TK",10,230,2,BLACK, BLACK,1);

ss5=true;
    }
}

```

//Section 6c: Counting the number of items selected in Main Dish

```

if(ss1==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{s1=s1+1;}

    else if(is_pressed(5, 400, 90, 435,p.x,p.y) && s1>0)
{s1=s1-1;}
}

if(ss2==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{s2=s2+1;}

    else if(is_pressed(5, 400, 90, 435,p.x,p.y) && s2>0)
{s2=s2-1;}
}

if(ss3==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{s3=s3+1;}

    else if(is_pressed(5, 400, 90, 435,p.x,p.y) && s3>0)
{s3=s3-1;}
}

if(ss4==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{s4=s4+1;}

    else if(is_pressed(5, 400, 90, 435,p.x,p.y) && s4>0)
{s4=s4-1;}
}

if(ss5==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{s5=s5+1;}

    else if(is_pressed(5, 400, 90, 435,p.x,p.y) && s5>0)
{s5=s5-1;}
}

```

```

}
}

```

```

//Section 7a: Selecting Appetizer and unselecting menu by pressing:

```

```

if(is_pressed(5,39,my_lcd.Get_Display_Width()-1,57,p.x,p.y)) //boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
{
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);

    switch_flag_2=true;
    switch_flag_3=true;
    switch_flag_4=true;
    switch_flag_6=true;

    my_lcd.Set_Draw_color(CREAM);
    my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 137);
    show_string(" ",5,119,2,BLUE, BLACK,1);
    show_string("Appetizer >",5,40,2,BLUE, BLACK,1);
    show_string("Drinks >",5,60,2,BLUE, BLACK,1);
    show_string("Kids menu >",5,80,2,BLUE, BLACK,1);
    show_string("Dessert >",5,100,2,BLUE, BLACK,1);
    show_string("Main Dish >",5,119,2,BLUE, BLACK,1);

    if(switch_flag_5)
    {
        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 57);
        show_string(" ",5,39,2,BLUE, BLACK,1);
        show_string("Appetizer <",5,39,2,BLUE, BLACK,1);

        Appetizer_menu();
        switch_flag_5 = false; //Selected
        //Show sppetizer pic, change it // show appetizer pic show_pic2
    }
    else
    {
        show_string(" ",5,39,2,BLUE, BLACK,1);
        show_string("Appetizer >",5,39,2,BLUE, BLACK,1);
        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);
        switch_flag_5 = true; //Not selected
    }
}

//Selecting Chicken Wings and unselect all
if(switch_flag_5==false)
{
    if(is_pressed(0,147,my_lcd.Get_Display_Width()-1,166,p.x,p.y)){
        if(dd1)
        {
            my_lcd.Set_Draw_color(BLUE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
            show_string("Chicken Wings 150TK",13,150,2,BLACK, BLACK,1);
            menu_flag2 = false;
            ddl=false;
            show_pic();
            my_lcd.Set_Draw_color(LIGHTYELLOW);
            my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
            show_string(" ",10,270,1,BLACK, BLACK,1);
            show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
            show_string("chicken wings,Vegetable oil,garlic powder,Kosher salt,",10,290,1,BLACK, BLACK,1);
            show_string("black pepper,hot sauce,honey,butter",10,310,1,BLACK, BLACK,1);

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
            show_string(" ",13,170,2,BLACK, BLACK,1);
            show_string("Pizza Fries 150TK",13,170,2,BLACK, BLACK,1);
            dd2 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
            show_string("Rice Ball 120TK",13,190,2,BLACK, BLACK,1);
            dd3 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
            show_string("Beef Patty 100TK",13,210,2,BLACK, BLACK,1);
            dd4 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Broccoli Rabe 70TK",13,230,2,BLACK, BLACK,1);
            dd5 = true;
        }
        else {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
            show_string("Chicken Wings 150TK",13,150,2,BLACK, BLACK,1);
            dd1 = true;
        }
    }
}

else if(is_pressed(0,167,my_lcd.Get_Display_Width()-1,186,p.x,p.y)){
    if(dd2)
    {
        my_lcd.Set_Draw_color(BLUE);
        my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
        show_string("Pizza Fries 150TK",10,170,2,BLACK, BLACK,1);
    }
}

```

```

dd2 = false;
show_pic2();
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("frozen shoestring fries, pepperoni slices,ground sausage,",10,290,1,BLACK, BLACK,1);
show_string("pizza sauce,banana peppers,mozzarella,Parmesan",10,310,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Chicken Wings 150TK",10,150,2,BLACK, BLACK,1);
ddl = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Rice Ball 120TK",10,190,2,BLACK, BLACK,1);
dd3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Beef Patty 100TK",10,210,2,BLACK, BLACK,1);
dd4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Broccoli Rabe 70TK",10,230,2,BLACK, BLACK,1);
dd5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Pizza Fries 150TK",10,170,2,BLACK, BLACK,1);
dd2=true;
}
}

else if (is_pressed(0,187,my_lcd.Get_Display_Width()-1,206,p.x,p.y)) {
if (dd3)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Rice Ball 120TK",10,190,2,BLACK, BLACK,1);

dd3=false;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Chicken Wings 150TK",10,150,2,BLACK, BLACK,1);
ddl = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Pizza Fries 150TK",10,170,2,BLACK, BLACK,1);
dd2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Beef Patty 100TK",10,210,2,BLACK, BLACK,1);
dd4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Broccoli Rabe 70TK",10,230,2,BLACK, BLACK,1);
dd5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Rice Ball 120TK",10,190,2,BLACK, BLACK,1);

dd3=true;
}
}

else if (is_pressed(0,207,my_lcd.Get_Display_Width()-1,226,p.x,p.y)) {
if (dd4)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Beef Patty 100TK",10,210,2,BLACK, BLACK,1);
show_pic3();
dd4=false;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Chicken Wings 150TK",10,150,2,BLACK, BLACK,1);
ddl = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Pizza Fries 150TK",10,170,2,BLACK, BLACK,1);
dd2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Rice Ball 120TK",10,190,2,BLACK, BLACK,1);
dd3 = true;

my_lcd.Set_Draw_color(WHITE);

```

```

my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Broccoli Rabe      70TK",10,230,2,BLACK, BLACK,1);
dd5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Beef Patty        100TK",10,210,2,BLACK, BLACK,1);

dd4=true;
}

}

else if(is_pressed(0,227,my_lcd.Get_Display_Width()-1,246,p.x,p.y))
{
if(dd5)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Broccoli Rabe      70TK",10,230,2,BLACK, BLACK,1);

dd5=false;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Chicken Wings     150TK",10,150,2,BLACK, BLACK,1);
dd1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 186);
show_string("Pizza Fries       150TK",10,170,2,BLACK, BLACK,1);
dd2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Rice Ball         120TK",10,190,2,BLACK, BLACK,1);
dd3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Beef Patty        100TK",10,210,2,BLACK, BLACK,1);
dd4 = true;

}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Broccoli Rabe      70TK",10,230,2,BLACK, BLACK,1);

dd5=true;
}
}

//Counting the number of items selected in Main Dish
if(dd1==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{d1=d1+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && d1>0)
{d1=d1-1;}
}

if(dd2==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{d2=d2+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && d2>0)
{d2=d2-1;}
}

if(dd3==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{d3=d3+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && d3>0)
{d3=d3-1;}
}

if(dd4==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{d4=d4+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && d4>0)
{d4=d4-1;}
}

if(dd5==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{d5=d5+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && d5>0)
{d5=d5-1;}
}

}
}

//Selecting Kids menu by touch
if(is_pressed(5,78,my_lcd.Get_Display_Width()-1,98,p.x,p.y)) //boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
{
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);

```

```

switch_flag_2=true;
switch_flag_3=true;
switch_flag_5=true;
switch_flag_6=true;

my_lcd.Set_Draw_color(CREAM);
my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 137);
show_string(" ",5,119,2,BLUE, BLUE,1);
show_string("Appetizer >",5,40,2,BLUE, BLACK,1);
show_string("Drinks >",5,60,2,BLUE, BLACK,1);
show_string("Kids menu >",5,80,2,BLUE, BLACK,1);
show_string("Desert >",5,100,2,BLUE, BLACK,1);
show_string("Main Dish >",5,119,2,BLUE, BLACK,1);

if(switch_flag_4)
{
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 78, my_lcd.Get_Display_Width()-140, 98);
    show_string(" ",5,80,2,BLUE, BLACK,1);
    show_string("Kids menu <",5,80,2,BLUE, BLACK,1);

    Kids_menu();
    switch_flag_4 = false; //Selected
}
else
{
    show_string(" ",5,80,2,BLUE, BLACK,1);
    show_string("Kids menu >",5,80,2,BLUE, BLACK,1);
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);
    switch_flag_4 = true; //Not selected
}
}
//If Kids menu is selected:

if(switch_flag_4==false)
{
    //Select Cheese Sandwich and Unselect all
    if(is_pressed(0,147,my_lcd.Get_Display_Width()-1,166,p.x,p.y)){
        if(ff1)
        {
            my_lcd.Set_Draw_color(BLUE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
            show_string("Cheese Sandwich 100TK",13,150,2,BLACK, BLACK,1);
            ff1=false;
            show_pic1();
            my_lcd.Set_Draw_color(LIGHTYELLOW);
            my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
            show_string(" ",10,270,1,BLACK, BLACK,1);
            show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
            show_string("White bread, butter, cheddar cheese",10,290,1,BLACK, BLACK,1);

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
            show_string(" ",13,170,2,BLACK, BLACK,1);
            show_string("Hamburger 140TK",13,170,2,BLACK, BLACK,1);
            ff2 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
            show_string("Steak Fingers 120TK",13,190,2,BLACK, BLACK,1);
            ff3 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
            show_string("Chicken Fingers 120TK",13,210,2,BLACK, BLACK,1);
            ff4 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Beef Steak 160TK",13,230,2,BLACK, BLACK,1);
            ff5 = true;
        }
        else {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
            show_string("Cheese Sandwich 100TK",13,150,2,BLACK, BLACK,1);
            ff1 = true;
        }
    }
}

//Select Spicy food and Unselect all
else if(is_pressed(0,167,my_lcd.Get_Display_Width()-1,186,p.x,p.y)){
    if(ff2)
    {
        my_lcd.Set_Draw_color(BLUE);
        my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
        show_string("Hamburger 140TK",10,170,2,BLACK, BLACK,1);
        ff2 = false;

        show_pic2();
        my_lcd.Set_Draw_color(LIGHTYELLOW);
        my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
        show_string(" ",10,270,1,BLACK, BLACK,1);
        show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
        show_string("lean ground beef,onion,Cheddar cheese,soy sauce,Worcestershire sauce",10,290,1,BLACK, BLACK,1);
        show_string("egg,garlic,parsley,basil,dried oregano,crushed dried rosemary,salt",10,310,1,BLACK, BLACK,1);

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
        show_string("Cheese Sandwich 100TK",10,150,2,BLACK, BLACK,1);
        ff1 = true;
    }
}

```



```

        my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Steak Fingers    120TK",10,190,2,BLACK, BLACK,1);
ff3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);
ff4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Beef Steak      160TK",10,230,2,BLACK, BLACK,1);
ff5 = true;
    }
    else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Hamburger      140TK",10,170,2,BLACK, BLACK,1);
ff2=true;
    }
}

//Select Pizza and Unselect all
else if (is_pressed(0,187,my_lcd.Get_Display_Width()-1,206,p.x,p.y)){

if (ff3)
{
    my_lcd.Set_Draw_color(BLUE);
    my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
    show_string("Steak Fingers    120TK",10,190,2,BLACK, BLACK,1);

    ff3=false;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Cheese Sandwich  100TK",10,150,2,BLACK, BLACK,1);
ff1 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Hamburger      140TK",10,170,2,BLACK, BLACK,1);
ff2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);
ff4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Beef Steak      160TK",10,230,2,BLACK, BLACK,1);
ff5 = true;
    }
    else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Steak Fingers    120TK",10,190,2,BLACK, BLACK,1);

ff3=true;
    }
}

//Select Spaghetti and Unselect all
else if (is_pressed(0,207,my_lcd.Get_Display_Width()-1,226,p.x,p.y)){

if (ff4)
{
    my_lcd.Set_Draw_color(BLUE);
    my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
    show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);
    show_pic3();
    ff4=false;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string("Cheese Sandwich  100TK",10,150,2,BLACK, BLACK,1);
ff1 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Hamburger      140TK",10,170,2,BLACK, BLACK,1);
ff2 = true;

    my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Steak Fingers    120TK",10,190,2,BLACK, BLACK,1);
ff3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Beef Steak      160TK",10,230,2,BLACK, BLACK,1);
ff5 = true;
    }
    else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);

ff4=true;
    }
}

```

```

    }

//Select Pasta and Unselect all

    else if (is_pressed(0,227,my_lcd.Get_Display_Width()-1,246,p.x,p.y))
    {
        if (ff5)
        {
            my_lcd.Set_Draw_color(BLUE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Beef Steak      160TK",10,230,2,BLACK, BLACK,1);

            ff5=false;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
            show_string("Cheese Sandwich 100TK",10,150,2,BLACK, BLACK,1);
            ff1 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
            show_string("Hamburger      140TK",10,170,2,BLACK, BLACK,1);
            ff2 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
            show_string("Steak Fingers  120TK",10,190,2,BLACK, BLACK,1);
            ff3 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
            show_string("Chicken Fingers 120TK",10,210,2,BLACK, BLACK,1);
            ff4 = true;

        }
        else {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Beef Steak      160TK",10,230,2,BLACK, BLACK,1);

            ff5=true;
        }
    }

//Counting the number of items selected in Main Dish
    if (ff1==false)
    {
        if (is_pressed(95, 400,160, 435,p.x,p.y))
        {
            f1=f1+1;

            else if (is_pressed(5, 400, 90, 435,p.x,p.y) && f1>0)
            {
                f1=f1-1;
            }

        }

        if (ff2==false)
        {
            if (is_pressed(95, 400,160, 435,p.x,p.y))
            {
                f2=f2+1;

                else if (is_pressed(5, 400, 90, 435,p.x,p.y) && f2>0)
                {
                    f2=f2-1;
                }

            }

            if (ff3==false)
            {
                if (is_pressed(95, 400,160, 435,p.x,p.y))
                {
                    f3=f3+1;

                    else if (is_pressed(5, 400, 90, 435,p.x,p.y) && f3>0)
                    {
                        f3=f3-1;
                    }

                }

                if (ff4==false)
                {
                    if (is_pressed(95, 400,160, 435,p.x,p.y))
                    {
                        f4=f4+1;

                        else if (is_pressed(5, 400, 90, 435,p.x,p.y) && f4>0)
                        {
                            f4=f4-1;
                        }

                    }

                    if (ff5==false)
                    {
                        if (is_pressed(95, 400,160, 435,p.x,p.y))
                        {
                            f5=f5+1;

                            else if (is_pressed(5, 400, 90, 435,p.x,p.y) && f5>0)
                            {
                                f5=f5-1;
                            }

                        }

                    }

                }

            }

//Selecting Desert menu by touch
            if (is_pressed(5,98,my_lcd.Get_Display_Width()-1,117,p.x,p.y)) //boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
            {
                my_lcd.Set_Draw_color(WHITE);
                my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);

                switch_flag_2=true;
                switch_flag_5=true;
                switch_flag_4=true;
                switch_flag_6=true;

                my_lcd.Set_Draw_color(CREAM);
                my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 137);
                show_string(" ",5,100,2,BLUE, BLUE,1);
                show_string("Appetizer  >",5,40,2,BLUE, BLACK,1);
                show_string("Drinks    >",5,60,2,BLUE, BLACK,1);
                show_string("Kids menu  >",5,80,2,BLUE, BLACK,1);
                show_string("Dessert    >",5,100,2,BLUE, BLACK,1);
            }
        }
    }

```

```

show_string("Main Dish  >",5,119,2,BLUE, BLACK,1);

if(switch_flag_3)
{
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 98, my_lcd.Get_Display_Width()-140, 117);
    show_string(" ",5,119,2,BLUE, BLACK,1);
    show_string("Dessert    <",5,100,2,BLUE, BLACK,1);

    Dessert_menu();
    switch_flag_3 = false; //Selected
}
else
{
    show_string(" ",5,100,2,BLUE, BLACK,1);
    show_string("Dessert    >",5,100,2,BLUE, BLACK,1);
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);
    switch_flag_3 = true; //Not selected
}
}

//If Desert menu is selected:

if(switch_flag_3==false)
{
    //Select Lava Cake and Unselect all
    if(is_pressed(0,147,my_lcd.Get_Display_Width()-1,166,p.x,p.y)){
        if(gg1)
        {
            my_lcd.Set_Draw_color(BLUE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
            show_string("Lava Cake      120TK",13,150,2,BLACK, BLACK,1);
            menu_flag1 = false;
            gg1=false;
            show_pic();
            my_lcd.Set_Draw_color(LIGHTYELLOW);
            my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
            show_string("Cheesecake      130TK",13,170,2,BLACK, BLACK,1);
            gg2 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
            show_string("Maple Apple Pie  130TK",13,190,2,BLACK, BLACK,1);
            gg3 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
            show_string("Brownie Pops      90TK",13,210,2,BLACK, BLACK,1);
            gg4 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Cherry pie      120TK",13,230,2,BLACK, BLACK,1);
            gg5 = true;
        }
        else {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
            show_string("Lava Cake      120TK",13,150,2,BLACK, BLACK,1);
            gg1 = true;
        }
    }
}

//Select Cheese cake and Unselect all
else if(is_pressed(0,167,my_lcd.Get_Display_Width()-1,186,p.x,p.y)){
    if(gg2)
    {
        my_lcd.Set_Draw_color(BLUE);
        my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
        show_string("Cheesecake      130TK",10,170,2,BLACK, BLACK,1);
        gg2 = false;
        show_pic2();
        my_lcd.Set_Draw_color(LIGHTYELLOW);
        my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
        show_string("Lava Cake      120TK",10,150,2,BLACK, BLACK,1);
        gg1 = true;

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
        show_string("Maple Apple Pie  130TK",10,190,2,BLACK, BLACK,1);
        gg3 = true;

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
        show_string("Brownie Pops      90TK",10,210,2,BLACK, BLACK,1);
        gg4 = true;

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
        show_string("Cherry pie      120TK",10,230,2,BLACK, BLACK,1);
        gg5 = true;
    }
    else {
        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
    }
}

```

```

show_string("Cheesecake      130TK",10,170,2,BLACK, BLACK,1);
gg2=true;
}
}

//Select Pizza and Unselect all
else if (is_pressed(0,187,my_lcd.Get_Display_Width()-1,206,p.x,p.y)){

if (gg3)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Maple Apple Pie 130TK",10,190,2,BLACK, BLACK,1);

gg3=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("All-purpose flour, salt, shortening, tart apples, brown sugar,",10,290,1,BLACK, BLACK,1);
show_string("margarine, maple-flavored syrup",10,310,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Lava Cake      120TK",10,150,2,BLACK, BLACK,1);
gg1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Cheesecake      130TK",10,170,2,BLACK, BLACK,1);
gg2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Brownie Pops      90TK",10,210,2,BLACK, BLACK,1);
gg4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Cherry pie      120TK",10,230,2,BLACK, BLACK,1);
gg5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Maple Apple Pie 130TK",10,190,2,BLACK, BLACK,1);

gg3=true;
}
}

//Select Spaghetti and Unselect all
else if (is_pressed(0,207,my_lcd.Get_Display_Width()-1,226,p.x,p.y)){

if (gg4)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Brownie Pops      90TK",10,210,2,BLACK, BLACK,1);
show_pic3();
gg4=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("fudge brownie mix, vegetable oil, water, eggs,",10,290,1,BLACK, BLACK,1);
show_string("dark cocoa candy melts,candy sprinkles",10,310,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Lava Cake      120TK",10,150,2,BLACK, BLACK,1);
gg1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Cheesecake      130TK",10,170,2,BLACK, BLACK,1);
gg2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Maple Apple Pie 130TK",10,190,2,BLACK, BLACK,1);
gg3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Cherry pie      120TK",10,230,2,BLACK, BLACK,1);
gg5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Brownie Pops      90TK",10,210,2,BLACK, BLACK,1);

gg4=true;
}
}

```

```
//Select Pasta and Unselect all
```

```
else if(is_pressed(0,227,my_lcd.Get_Display_Width()-1,246,p.x,p.y))
{if(gg5)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Cherry pie 120TK",10,230,2,BLACK, BLACK,1);

gg5=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("frozen tart cherries, granulated sugar, cornstarch",10,290,1,BLACK, BLACK,1);
show_string("almond extract, pie crust, butter, granulated sugar",10,310,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Lava Cake 120TK",10,150,2,BLACK, BLACK,1);
gg1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Cheesecake 130TK",10,170,2,BLACK, BLACK,1);
gg2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Maple Apple Pie 130TK",10,190,2,BLACK, BLACK,1);
gg3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Brownie Pops 90TK",10,210,2,BLACK, BLACK,1);
gg4 = true;

}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Cherry pie 120TK",10,230,2,BLACK, BLACK,1);

gg5=true;
}
}
```

```
//Counting the number of items selected in Main Dish
```

```
if(gg1==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{g1=g1+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && g1>0)
{g1=g1-1;}
}

if(gg2==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{g2=g2+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && g2>0)
{g2=g2-1;}
}

if(gg3==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{g3=g3+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && g3>0)
{g3=g3-1;}
}

if(gg4==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{g4=g4+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && g4>0)
{g4=g4-1;}
}

if(gg5==false)
{ if(is_pressed(95, 400,160, 435,p.x,p.y))
{g5=g5+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && g5>0)
{g5=g5-1;}
}
}
```

```
//Selecting Drinks menu by touch
```

```
if(is_pressed(5,58,my_lcd.Get_Display_Width()-1,78,p.x,p.y)) //boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py)
{
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);

switch_flag_5=true;
switch_flag_3=true;
switch_flag_4=true;
switch_flag_6=true;

my_lcd.Set_Draw_color(CREAM);
my_lcd.Fill_Rectangle(0, 37, my_lcd.Get_Display_Width()-140, 137);
show_string(" ",5,60,2,BLUE, BLUE,1);
```

```

show_string("Appetizer   >",5,40,2,BLUE, BLACK,1);
show_string("Drinks     >",5,60,2,BLUE, BLACK,1);
show_string("Kids menu  >",5,80,2,BLUE, BLACK,1);
show_string("Dessert    >",5,100,2,BLUE, BLACK,1);
show_string("Main Dish  >",5,119,2,BLUE, BLACK,1);

if (switch_flag_2)
{
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 58, my_lcd.Get_Display_Width()-140, 78);
    show_string(" ",5,60,2,BLUE, BLACK,1);
    show_string("Drinks     <",5,60,2,BLUE, BLACK,1);

    Drinks_menu();
    switch_flag_2 = false; //Selected
}
else
{
    show_string(" ",5,60,2,BLUE, BLACK,1);
    show_string("Drinks     >",5,60,2,BLUE, BLACK,1);
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 250);
    switch_flag_2 = true; //Not selected
}
}

//If Drinks menu is selected:

if (switch_flag_2==false)
{
    //Select Fried Rice and Unselect all
    if (is_pressed(0,147,my_lcd.Get_Display_Width()-1,166,p.x,p.y)){
        if (hh1)
        {
            my_lcd.Set_Draw_color(BLUE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 166);
            show_string("Cappuccino      100TK",13,150,2,BLACK, BLACK,1);
            menu_flag1 = false;
            hh1=false;
            show_pic();
            my_lcd.Set_Draw_color(LIGHTYELLOW);
            my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

            show_string(" ",10,270,1,BLACK, BLACK,1);
            show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
            show_string("finely ground dark roast coffee, water, foamed milk",10,290,1,BLACK, BLACK,1);

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
            show_string(" ",13,170,2,BLACK, BLACK,1);
            show_string("Glace      80TK",13,170,2,BLACK, BLACK,1);
            hh2 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
            show_string("Latte      100TK",13,190,2,BLACK, BLACK,1);
            hh3 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
            show_string("Cocoa      120TK",13,210,2,BLACK, BLACK,1);
            hh4 = true;

            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
            show_string("Mocha      120TK",13,230,2,BLACK, BLACK,1);
            hh5 = true;
        }
        else {
            my_lcd.Set_Draw_color(WHITE);
            my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
            show_string("Cappuccino      100TK",13,150,2,BLACK, BLACK,1);
            hh1 = true;
        }
    }
}

//Select Spicy food and Unselect all
else if (is_pressed(0,167,my_lcd.Get_Display_Width()-1,186,p.x,p.y)){
    if (hh2)
    {
        my_lcd.Set_Draw_color(BLUE);
        my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
        show_string("Glace      80TK",10,170,2,BLACK, BLACK,1);
        hh2 = false;
        my_lcd.Set_Draw_color(LIGHTYELLOW);
        my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
        show_string(" ",10,270,1,BLACK, BLACK,1);
        show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
        show_string("dried thyme, fresh parsley stems, whole black peppercorns",10,290,1,BLACK, BLACK,1);
        show_string("bay leaf, brown sauce (Espagnole sauce),beef stock",10,310,1,BLACK, BLACK,1);

        show_pic2();

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
        show_string(" ",10,150,2,BLACK, BLACK,1);
        show_string("Cappuccino      100TK",10,150,2,BLACK, BLACK,1);
        hh1 = true;

        my_lcd.Set_Draw_color(WHITE);
        my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
    }
}

```

```

show_string("Latte          100TK",10,190,2,BLACK, BLACK,1);
hh3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Cocoa          120TK",10,210,2,BLACK, BLACK,1);
hh4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Mocha          120TK",10,230,2,BLACK, BLACK,1);
hh5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Glace          80TK",10,170,2,BLACK, BLACK,1);
hh2=true;
}
}

//Select Pizza and Unselect all
else if (is_pressed(0,187,my_lcd.Get_Display_Width()-1,206,p.x,p.y)){

if (hh3)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Latte          100TK",10,190,2,BLACK, BLACK,1);

hh3=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("milk, freshly brewed dark roast espresso coffee",10,290,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Cappuccino      100TK",10,150,2,BLACK, BLACK,1);
hh1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Glace          80TK",10,170,2,BLACK, BLACK,1);
hh2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Cocoa          120TK",10,210,2,BLACK, BLACK,1);
hh4 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Mocha          120TK",10,230,2,BLACK, BLACK,1);
hh5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Latte          100TK",10,190,2,BLACK, BLACK,1);

hh3=true;
}
}

//Select Spaghetti and Unselect all
else if (is_pressed(0,207,my_lcd.Get_Display_Width()-1,226,p.x,p.y)){

if (hh4)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Cocoa          120TK",10,210,2,BLACK, BLACK,1);
show_pic3();
hh4=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);
show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("unsweetened cocoa powder, sugar, salt",10,290,1,BLACK, BLACK,1);
show_string("milk, vanilla extract",10,310,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Cappuccino      100TK",10,150,2,BLACK, BLACK,1);
hh1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Glace          80TK",10,170,2,BLACK, BLACK,1);
hh2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Latte          100TK",10,190,2,BLACK, BLACK,1);
hh3 = true;
}
}

```

```

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Mocha          120TK",10,230,2,BLACK, BLACK,1);
hh5 = true;
}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Cocoa          120TK",10,210,2,BLACK, BLACK,1);

hh4=true;
}
}

//Select Pasta and Unselect all

else if(is_pressed(0,227,my_lcd.Get_Display_Width()-1,246,p.x,p.y))
{
if(hh5)
{
my_lcd.Set_Draw_color(BLUE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Mocha          120TK",10,230,2,BLACK, BLACK,1);

hh5=false;
my_lcd.Set_Draw_color(LIGHTYELLOW);
my_lcd.Fill_Rectangle(0, 260, my_lcd.Get_Display_Width()-1, 340);

show_string(" ",10,270,1,BLACK, BLACK,1);
show_string("Ingredients:",10,270,1,BLACK, BLACK,1);
show_string("ground espresso, milk, drinking chocolate",10,290,1,BLACK, BLACK,1);

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 166);
show_string(" ",10,150,2,BLACK, BLACK,1);
show_string("Cappuccino      100TK",10,150,2,BLACK, BLACK,1);
hh1 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 167, my_lcd.Get_Display_Width()-1, 186);
show_string("Glace          80TK",10,170,2,BLACK, BLACK,1);
hh2 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 187, my_lcd.Get_Display_Width()-1, 206);
show_string("Latte          100TK",10,190,2,BLACK, BLACK,1);
hh3 = true;

my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 207, my_lcd.Get_Display_Width()-1, 226);
show_string("Cocoa          120TK",10,210,2,BLACK, BLACK,1);
hh4 = true;

}
else {
my_lcd.Set_Draw_color(WHITE);
my_lcd.Fill_Rectangle(0, 227, my_lcd.Get_Display_Width()-1, 246);
show_string("Mocha          120TK",10,230,2,BLACK, BLACK,1);

hh5=true;
}
}

}

//Counting the number of items selected in Main Dish
if(hh1==false)
{
if(is_pressed(95, 400,160, 435,p.x,p.y))
{hh1=hh1+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && hh1>0)
{hh1=hh1-1;}
}

if(hh2==false)
{
if(is_pressed(95, 400,160, 435,p.x,p.y))
{hh2=hh2+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && hh2>0)
{hh2=hh2-1;}
}

if(hh3==false)
{
if(is_pressed(95, 400,160, 435,p.x,p.y))
{hh3=hh3+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && hh3>0)
{hh3=hh3-1;}
}

if(hh4==false)
{
if(is_pressed(95, 400,160, 435,p.x,p.y))
{hh4=hh4+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && hh4>0)
{hh4=hh4-1;}
}

if(hh5==false)
{
if(is_pressed(95, 400,160, 435,p.x,p.y))
{hh5=hh5+1;}

else if(is_pressed(5, 400, 90, 435,p.x,p.y) && hh5>0)
{hh5=hh5-1;}
}
}
}

```



```

//Temperature sensing
val = analogRead(15);
float mv = ( val/1024.0)*5000;
float cel = mv/10;

//Section 11: Showing recommendation using temperature sensor reading
if (cel>30)
{
  my_lcd.Set_Draw_color(GREY);
  my_lcd.Fill_Rectangle(0, 0, my_lcd.Get_Display_Width()-1, 25);
  show_string(" ", 10, 290, 2, GREEN, GREEN, 1);///prolly need to set.color
  show_string("Try our Apple Juice!",30,10,2,LIGHTYELLOW, LIGHTYELLOW, 1);
}
else
{
  my_lcd.Set_Draw_color(GREY);
  my_lcd.Fill_Rectangle(0, 0, my_lcd.Get_Display_Width()-1, 25);
  show_string(" ", 10, 290, 2, LIGHTYELLOW, LIGHTYELLOW, 1);///prolly need to set.color
  show_string("Try our Hot Coffee!",40,10,2,LIGHTYELLOW, LIGHTYELLOW,1);
}

//int t_item= s1+s2+s3+s4+s5;
//my_lcd.Set_Draw_color(GREY);
//my_lcd.Print_Number_Int(t_item, 265, 450, 4, ' ', 10);

//Section 12: Cart designing and showing the selected items
int t_amount= (150*s1)+(150*d1)+(100*f1)+(120*g5)+(120*h5); //total price of items selected
if(is_pressed(220, 443, my_lcd.Get_Display_Width()-1, 480, p.x, p.y))

{
  m=0;
  if(switch_flag_cart)
  {
    my_lcd.Set_Draw_color(GREY);
    my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width()-1, 380);
    show_string("Cart",15,157,1,WHITE, WHITE,1);

    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Draw_Fast_HLine(15, 182, my_lcd.Get_Display_Width()-15);
    my_lcd.Set_Draw_color(WHITE);
    my_lcd.Draw_Fast_HLine(15, 182, my_lcd.Get_Display_Width()-15);

    my_lcd.Draw_Fast_HLine(15, 340, my_lcd.Get_Display_Width()-15);
    show_string(" ",157, 350, 1,WHITE, WHITE,1);
    show_string("Total: ",157,350,1,WHITE, WHITE,1);
    my_lcd.Print_Number_Int(t_amount, 210, 350, 4, ' ', 10);

    if(s1>0)
    {
      my_lcd.Set_Draw_color(WHITE);
      show_string("Fried Rice x",20, 190, 1,WHITE, WHITE,1);
      my_lcd.Print_Number_Int(s1, 200, 190, 4, ' ', 10);
      while(m<=8)
      {m=m+8;}
    }
    if(d1>0)
    {
      my_lcd.Set_Draw_color(WHITE);
      show_string("Chichen Wings x",20, 190+m, 1,WHITE, WHITE,1);
      my_lcd.Print_Number_Int(d1, 200, 190+m, 4, ' ', 10);
      while(m <=16)
      {m=m+8;}
    }
    if(f1>0)
    {
      my_lcd.Set_Draw_color(WHITE);
      show_string("Cheese Burger x",20, 190+m, 1,WHITE, WHITE,1);
      my_lcd.Print_Number_Int(f1, 200, 190+m, 4, ' ', 10);
      while(m <=24)
      {m=m+8;}
    }
    if(g5>0)
    {
      my_lcd.Set_Draw_color(WHITE);
      show_string("Cherry Pie x",20, 190+m, 1,WHITE, WHITE,1);
      my_lcd.Print_Number_Int(g5, 200, 190+m, 4, ' ', 10);
      while(m <=32)
      {m=m+8;}
    }
    if(h5>0)
    {
      my_lcd.Set_Draw_color(WHITE);
      show_string("Mocha x",20, 190+m, 1,WHITE, WHITE,1);
      my_lcd.Print_Number_Int(h5, 200, 190+m, 4, ' ', 10);
      while(m <=40)
      {m=m+8;}
    }
  }

  switch_flag_cart= false;
}

else
{
  my_lcd.Set_Draw_color(CREAM);
  my_lcd.Fill_Rectangle(0, 147, my_lcd.Get_Display_Width(), 380);
  switch_flag_cart = true;
}
}

//Section 16: Serial Transmission from Arduino Mega to NodeMCU
String myString = String(token);
String myString2 = String(s1);
str = String("Token Number: ") +String(myString)+String(";")+String("Food items :")+String("Friend Rice x")+String(myString2);
espSerial.print(str);

```

```

//Section 13: Placing Order
if(is_pressed(my_lcd.Get_Display_Width()-155, 400,my_lcd.Get_Display_Width()-5, 435,p.x,p.y))
//boolean is_pressed(int16_t x1,int16_t y1,int16_t x2,int16_t y2,int16_t px,int16_t py
{

    my_lcd.Set_Draw_color(GREY); //Lower part color
    my_lcd.Fill_Rectangle(my_lcd.Get_Display_Width()-155, 400,my_lcd.Get_Display_Width()-5, 435);
    show_string("Order Placed!",my_lcd.Get_Display_Width()-150,410,2,LIGHTYELLOW, LIGHTYELLOW,1);
    my_lcd.Set_Draw_color(GREY); //Lower part color
    my_lcd.Fill_Rectangle(0, 443, 220, 480);

    Serial.print("Fried Rice:");
    Serial.println(s1);
    Serial.print("Chicken Wings: ");
    Serial.println(d1);
    Serial.print("Cheese Sandwich: ");
    Serial.println(f1);
    Serial.print("Cheery Pie:");
    Serial.println(g5);
    Serial.print("Mocha:");
    Serial.println(h5);

    show_string("Token Number: ", 10, 450,2, LIGHTYELLOW, LIGHTYELLOW, 1);

    my_lcd.Set_Text_colour (LIGHTYELLOW);
    show_string(" ", 16, 450,2, LIGHTYELLOW, LIGHTYELLOW, 1);
    my_lcd.Print_Number_Int(token, 160, 450, 4, ' ', 10);
    delay(3000);

    my_lcd.Set_Draw_color(GREY); //Lower part color
    my_lcd.Fill_Rectangle(0, 443, 220, 480);
    show_string("Status: New Order", 10, 450,2, LIGHTYELLOW, LIGHTYELLOW, 1);
    my_lcd.Set_Draw_color(GREY); //Lower part color
    my_lcd.Fill_Rectangle(my_lcd.Get_Display_Width()-155, 400, my_lcd.Get_Display_Width()-5, 435);
    show_string("Place Order",my_lcd.Get_Display_Width()-150,410,2,LIGHTYELLOW, LIGHTYELLOW,1);

//Section 14: Printing details
void print_orders(){
    printer.setSize('L'); //Large text size
    printer.println(F("Token Number="));
    if(s1>0)
    {
        printer.setSize('S')
        printer.print(F("Fried Rice x"));
        printer.println(s1);
    }

    if(d1>0)
    {

        printer.setSize('S')
        printer.print(F("Chicken Wings x"));
        printer.println(d1);
    }
    if(f1>0)
    {
        printer.setSize('S')
        printer.print(F("Cheese Burger x"));
        printer.println(f1);
    }
    if(g5>0)
    {
        printer.setSize('S')
        printer.print(F("Cherry Pie x"));
        printer.println(g5);
    }
    if(h5>0)
    {
        printer.setSize('S')
        printer.println(F("Mocha x"));
        printer.println(h5);
    }
    printer.feed(3);
    printer.sleep(); //telling printer to sleep
    delay(3000L); //sleep for 3 seconds
    printer.wake(); //waking printer before printing again
    printer.setDefault(); //restoring printer to default
}

s1=0, d1=0, f1=0, g5=0, h5=0;
token++;

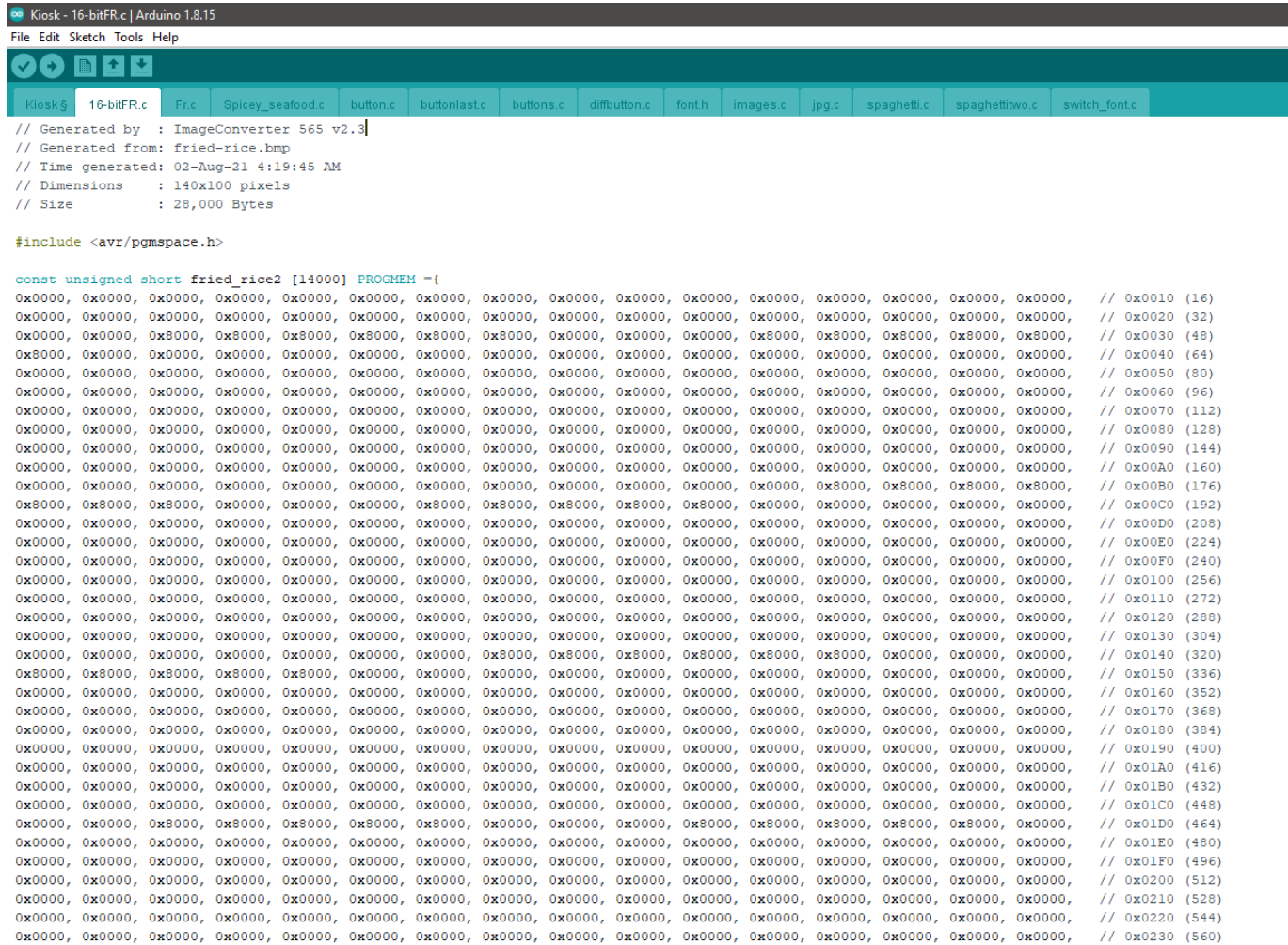
//Section 15: Sending signal to Seven Segment Display
//Blink
digitalWrite(31,HIGH);
digitalWrite(31,LOW);

}

}

```

C. .c file for Image



```
Kiosk - 16-bitFR.c | Arduino 1.8.15
File Edit Sketch Tools Help

Kiosk$ 16-bitFR.c F.c Spicely_seafood.c button.c buttonlast.c buttons.c diffbutton.c fon.h images.c jpg.c spaghetti.c spaghettiTwo.c switch_font.c

// Generated by : ImageConverter 565 v2.3
// Generated from: fried-rice.bmp
// Time generated: 02-Aug-21 4:19:45 AM
// Dimensions : 140x100 pixels
// Size : 28,000 Bytes

#include <avr/pgmspace.h>

const unsigned short fried_rice2 [14000] PROGMEM ={
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0010 (16)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0020 (32)
0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, // 0x0030 (48)
0x8000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0040 (64)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0050 (80)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0060 (96)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0070 (112)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0080 (128)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0090 (144)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x00A0 (160)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, // 0x00B0 (176)
0x8000, 0x8000, 0x8000, 0x0000, 0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x00C0 (192)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x00D0 (208)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x00E0 (224)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x00F0 (240)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0100 (256)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0110 (272)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0120 (288)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0130 (304)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0140 (320)
0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0150 (336)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0160 (352)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0170 (368)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0180 (384)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0190 (400)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x01A0 (416)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x01B0 (432)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x01C0 (448)
0x0000, 0x0000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x8000, 0x0000, // 0x01D0 (464)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x01E0 (480)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x01F0 (496)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0200 (512)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0210 (528)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0220 (544)
0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, // 0x0230 (560)
```

Fig. 16. .c file for Image